IGCSE 0607 Practice Questions — E7.5

Non-right-angled triangles

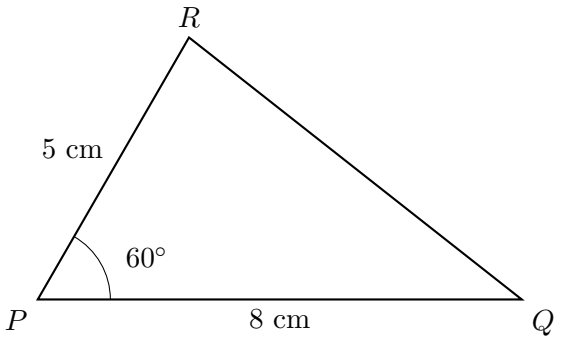
Total Questions: 39 — **Total Marks:** 280 Marks

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- **E7.5** Non-right-angled triangles
  → **Paper 2 (Non-Calculator)** *(3 Qs)* · **Paper 4 (Calculator)** *(36 Qs)*

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1 The diagram shows triangle PQR.
$PQ = 8$ cm, $PR = 5$ cm and angle $QPR = 60°$.



NOT TO SCALE

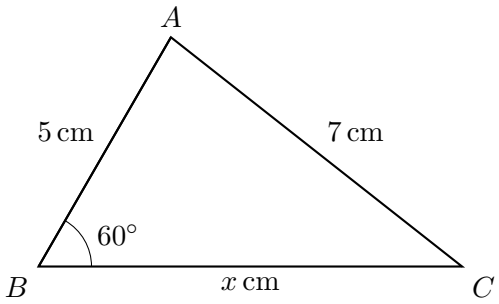
(a) Calculate the exact area of triangle PQR.
Give your answer in the form $a\sqrt{3}$.

..... cm^2 **[2]**

(b) Calculate the exact length of QR.

$QR =$ ..... cm **[3]**

2 In triangle ABC, $AB = 5\,\text{cm}$, $BC = x\,\text{cm}$ and $AC = 7\,\text{cm}$.
Angle $ABC = 60°$.



NOT TO SCALE

(a) Use the cosine rule to show that $x^2 - 5x - 24 = 0$.

[3]

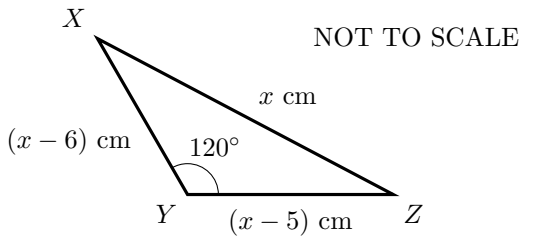
(b) Solve the equation $x^2 - 5x - 24 = 0$ to find the length of BC.

$BC =$ ..... cm **[2]**

(c) Calculate the exact area of triangle ABC.
Give your answer in the form $a\sqrt{b}$, where a and b are integers.

Area = ... cm^2 **[2]**

3 The diagram shows triangle XYZ with $XY = (x - 6)$ cm, $YZ = (x - 5)$ cm, $XZ = x$ cm, and angle $XYZ = 120°$.



(a) Use the cosine rule to show that $2x^2 - 33x + 91 = 0$.

[3]

(b) Solve the equation $2x^2 - 33x + 91 = 0$ and hence write down the lengths of XY, YZ, and XZ.

$XY =$ ..... , $YZ =$ ..... , $XZ =$ ..... **[3]**

(c) Calculate

(i) the exact area of triangle XYZ, giving your answer in the form $k\sqrt{3}$, where k is an integer,

Area = ... cm^2 **[2]**

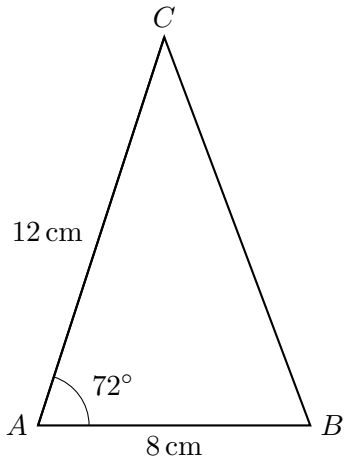
(ii) the exact shortest distance from vertex Y to the side XZ, giving your answer in the form $\frac{p\sqrt{3}}{q}$, where p and q are integers.

Distance = ... cm **[2]**

(d) Find the exact value of $\sin(\angle YXZ)$, giving your answer in the form $\dfrac{m\sqrt{3}}{n}$, where m and n are integers.

$\sin(\angle YXZ) =$ ..... **[2]**

4 The diagram shows triangle ABC.



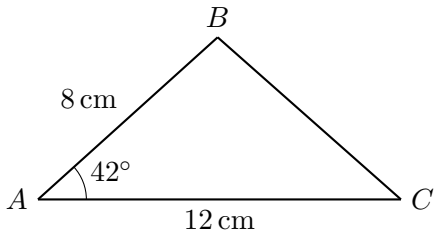
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

5 The diagram shows triangle ABC.
$AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 42°$.



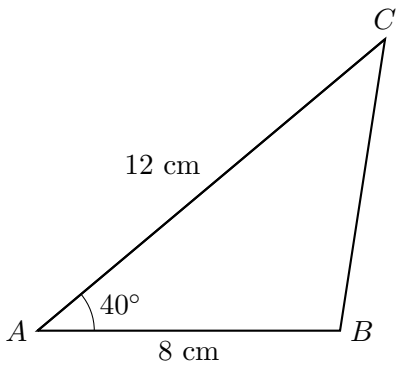
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

$BC =$ ..... cm **[3]**

6 The diagram shows a triangle ABC.



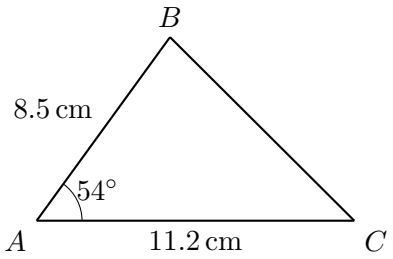
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

7 The diagram shows triangle ABC.



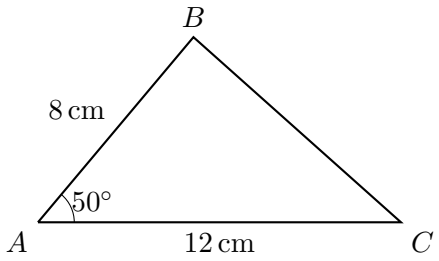
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

8 A triangle ABC is shown below.



NOT TO SCALE

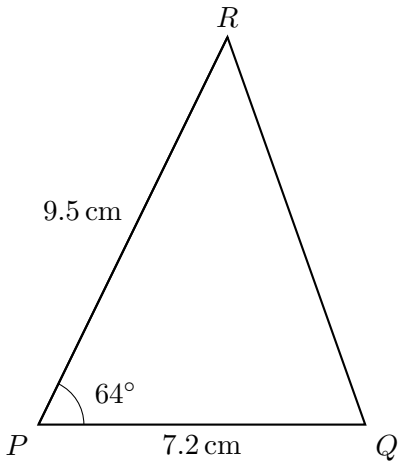
(a) Calculate the area of triangle ABC.

..... cm^2 **[2]**

(b) Calculate the length of BC.

..... cm **[3]**

9 In triangle PQR, $PQ = 7.2\,\text{cm}$, $PR = 9.5\,\text{cm}$ and angle $QPR = 64°$.



NOT TO SCALE

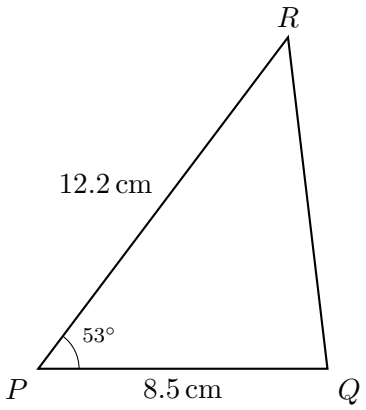
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

10 The diagram shows a triangle PQR.
$PQ = 8.5$ cm, $PR = 12.2$ cm and angle $QPR = 53°$.



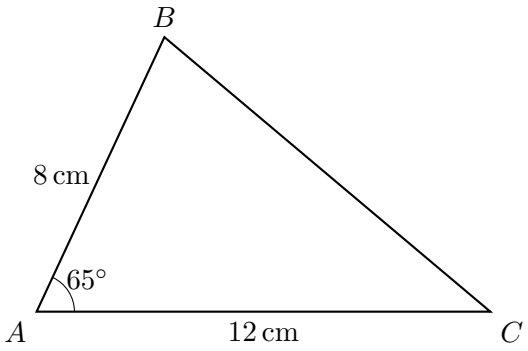
(a) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

(b) Calculate the length of QR.

QR = ... cm **[3]**

11 The diagram shows a triangle ABC with $AB = 8\,\text{cm}$, $AC = 12\,\text{cm}$ and angle $BAC = 65°$.



NOT TO SCALE

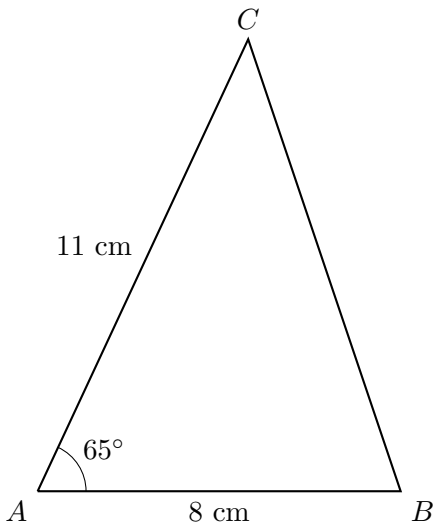
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

12 The diagram shows a triangle ABC.
$AB = 8$ cm, $AC = 11$ cm and angle $BAC = 65°$.



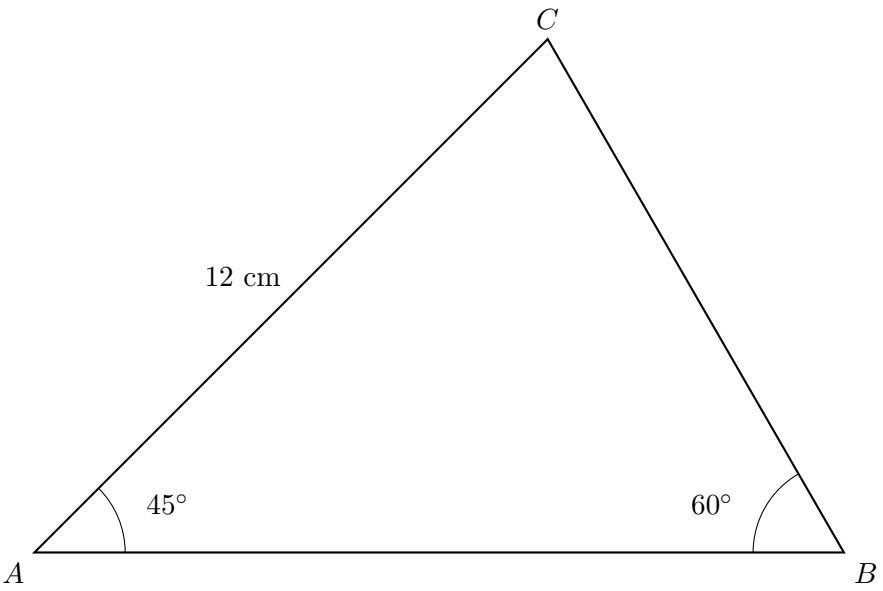
(a) Calculate the area of triangle ABC.

Area = ... cm^2 **[2]**

(b) Calculate the length of BC.

BC = ... cm **[3]**

13 In triangle ABC, $AC = 12$ cm, angle $BAC = 45°$ and angle $ABC = 60°$.



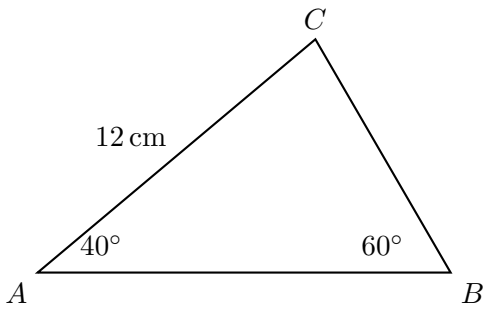
(a) Calculate the length of BC.

..... cm **[3]**

(b) Calculate the area of triangle ABC.

..... cm^2 **[3]**

14 The diagram shows triangle ABC.
$AC = 12$ cm, angle $BAC = 40°$ and angle $ABC = 60°$.



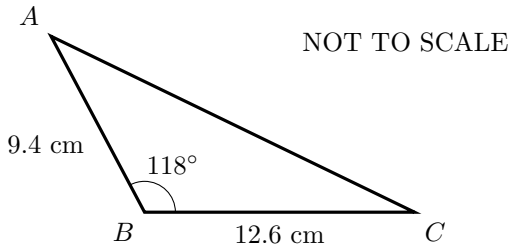
(a) Calculate the length of BC.

$BC =$ ..... cm **[3]**

(b) Calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

15 The diagram shows triangle ABC with $AB = 9.4$ cm, $BC = 12.6$ cm, and obtuse angle $ABC = 118°$.



(a) Calculate the length of AC, giving your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

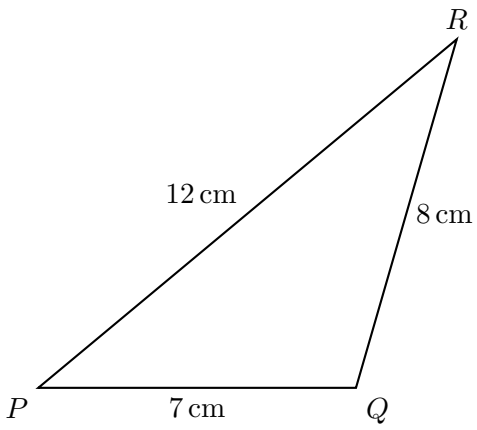
(b) Calculate angle BAC, giving your answer correct to 1 decimal place.

..... ° **[3]**

(c) Calculate the area of triangle ABC, giving your answer correct to 3 significant figures.

Area = ... cm^2 **[2]**

16 The diagram shows triangle PQR.
$PQ = 7$ cm, $QR = 8$ cm and $PR = 12$ cm.



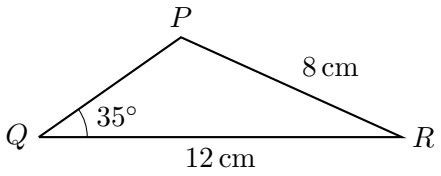
(a) Calculate the largest angle in triangle PQR.

..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[2]**

17 In triangle PQR, $PR = 8$ cm, $QR = 12$ cm and angle $PQR = 35°$.
Angle QPR is an obtuse angle.



NOT TO SCALE

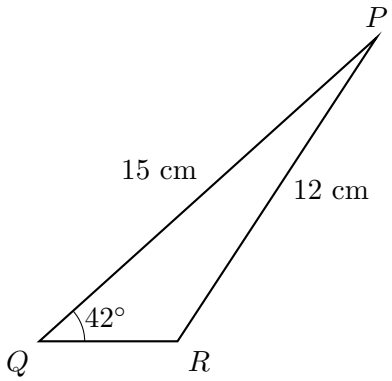
(a) Calculate the size of angle QPR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

18 The diagram shows a triangle PQR.
The length of PQ is 15 cm and the length of PR is 12 cm.
Angle $PQR = 42°$.
Angle PRQ is obtuse.



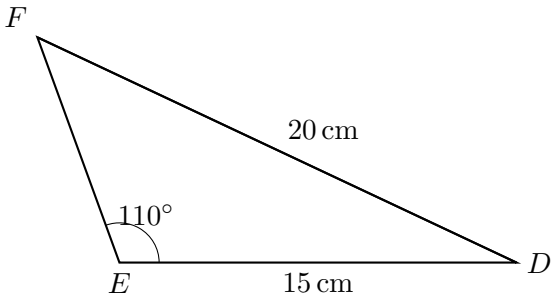
(a) Calculate angle PRQ.

..... ° **[3]**

(b) Calculate the length of QR.

..... cm **[3]**

19 The diagram shows triangle DEF.
$DE = 15$ cm, $DF = 20$ cm, and angle $DEF = 110°$.



NOT TO SCALE

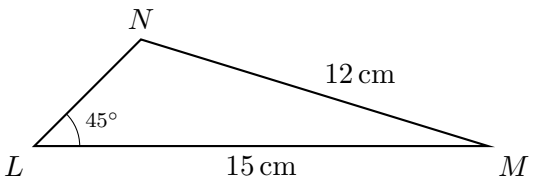
(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

20 The diagram shows a triangle LMN.
$LM = 15$ cm, $MN = 12$ cm and angle $MLN = 45°$.
Angle LNM is obtuse.



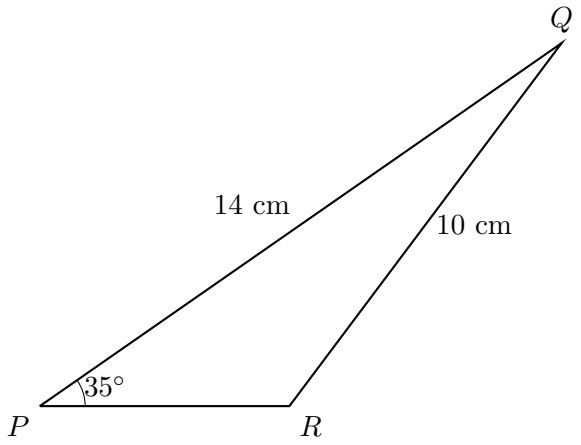
(a) Calculate the size of angle LNM.

Angle LNM = ... ° **[3]**

(b) Calculate the area of triangle LMN.

Area = ... cm^2 **[3]**

21 The diagram shows triangle PQR.
$PQ = 14$ cm, $QR = 10$ cm and angle $QPR = 35°$.
Angle PRQ is obtuse.



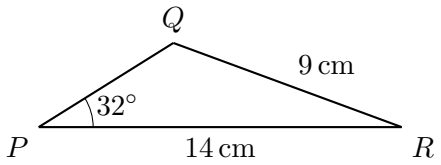
(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

(b) Calculate the area of triangle PQR.

Area = ... cm^2 **[3]**

22 The diagram shows triangle PQR.
The angle PQR is obtuse.



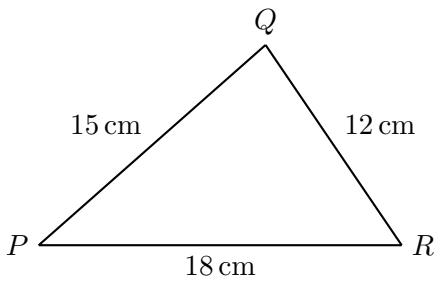
(a) Calculate the obtuse angle PQR.

..... ° **[4]**

(b) Calculate the area of triangle PQR.

..... cm^2 **[3]**

23 The diagram shows triangle PQR.
$PQ = 15\,\text{cm}$, $QR = 12\,\text{cm}$ and $PR = 18\,\text{cm}$.



(a) Use the cosine rule to show that angle $PQR = 82.8°$, correct to 1 decimal place.

[3]

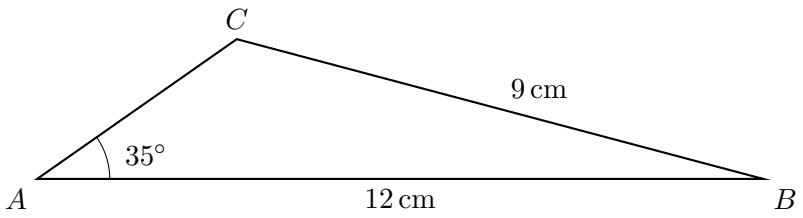
(b) Calculate the area of triangle PQR.

..... cm^2 **[2]**

(c) Calculate the shortest distance from R to the line PQ.

..... cm **[2]**

24 In triangle ABC, $AB = 12\,\text{cm}$, $BC = 9\,\text{cm}$ and angle $BAC = 35°$.



NOT TO SCALE

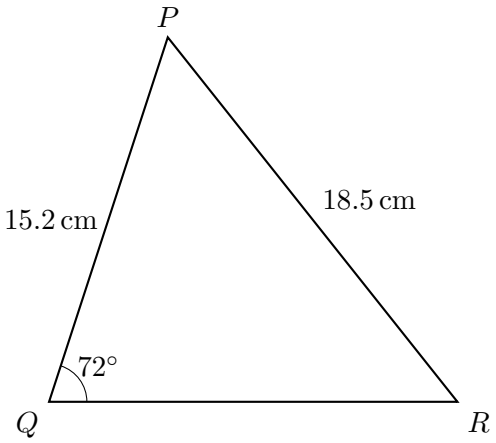
(a) Calculate the two possible values of angle ACB.
Give your answers correct to 1 decimal place.

..... ° **[4]**

(b) Given that angle ACB is obtuse, calculate the area of triangle ABC.

Area = ... cm^2 **[3]**

25 The diagram shows a triangle PQR.
$PQ = 15.2\,\text{cm}$, $PR = 18.5\,\text{cm}$ and angle $PQR = 72°$.



NOT TO SCALE

(a) Calculate the size of angle PRQ.

Angle $PRQ =$ ..... ° **[3]**

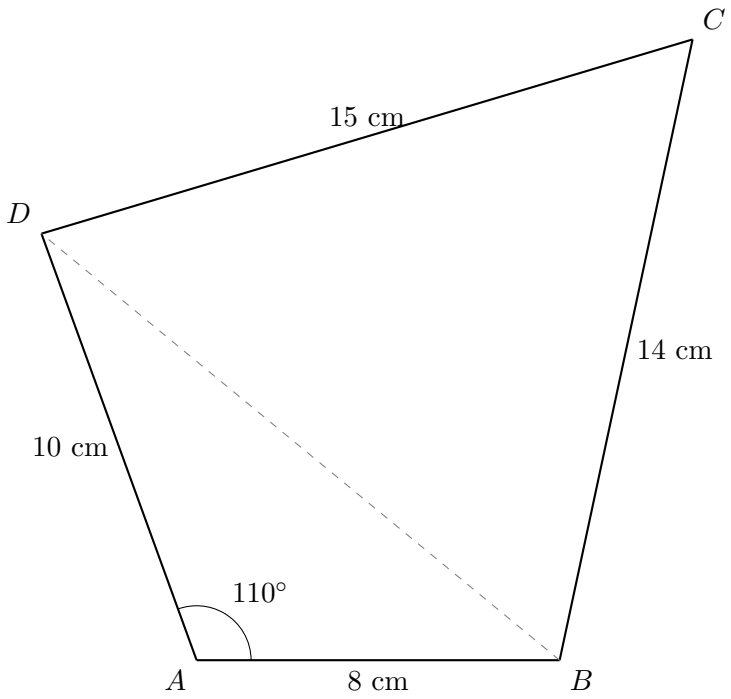
(b) Calculate the length of QR.

$QR =$ ..... cm **[3]**

(c) Calculate the shortest distance from P to the line QR.

Distance = ... cm **[2]**

26 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $AD = 10$ cm, $BC = 14$ cm and $CD = 15$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of the diagonal BD.

..... cm **[3]**

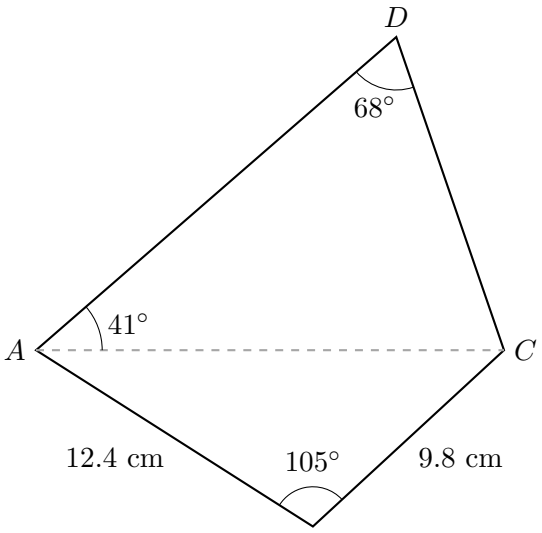
(b) Calculate the angle BCD.

..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

27 The diagram shows a quadrilateral $ABCD$.
$AB = 12.4$ cm, $BC = 9.8$ cm and angle $ABC = 105°$.
Angle $CAD = 41°$ and angle $ADC = 68°$.



NOT TO SCALE

(a) Calculate the length of AC.
Give your answer correct to 3 significant figures.

$AC =$ ..... cm **[3]**

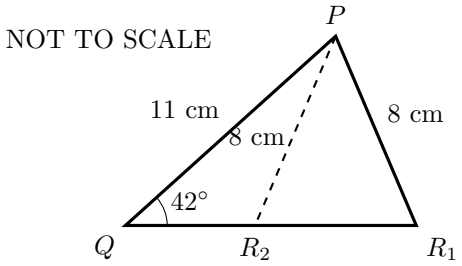
(b) Calculate the length of CD.
Give your answer correct to 3 significant figures.

$CD =$ ..... cm **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

28 In triangle PQR, $PQ = 11$ cm, $PR = 8$ cm, and angle $PQR = 42°$.
As shown in the diagram, there are two possible positions, R_1 and R_2, for vertex R.



(a) Calculate the two possible sizes of angle PRQ, giving your answers correct to 1 decimal place.

Acute angle = ... , Obtuse angle = ... **[4]**

(b) Write down the size of angle QPR_1 and the size of angle QPR_2, giving your answers correct to 1 decimal place.

$AngleQPR_1$ = ... , $AngleQPR_2$ = ... **[2]**

(c) Calculate the difference between the area of triangle PQR_1 and the area of triangle PQR_2, giving your answer correct to 3 significant figures.

Difference = ... cm^2 **[4]**

29 In triangle ABC, $AB = 15$ cm, $BC = 12$ cm and angle $BAC = 48°$.

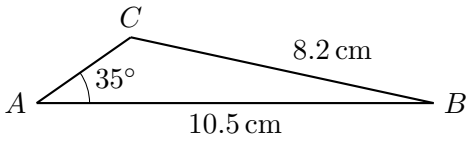
(a) Calculate the two possible values of angle BCA.

and ..... **[3]**

(b) Hence, calculate the difference between the two possible areas of triangle ABC.

..... cm^2 **[4]**

30 In triangle ABC, $AB = 10.5\,\text{cm}$, $BC = 8.2\,\text{cm}$ and angle $BAC = 35°$.
Angle BCA is obtuse.



(a) Calculate the size of angle BCA.

..... ° **[3]**

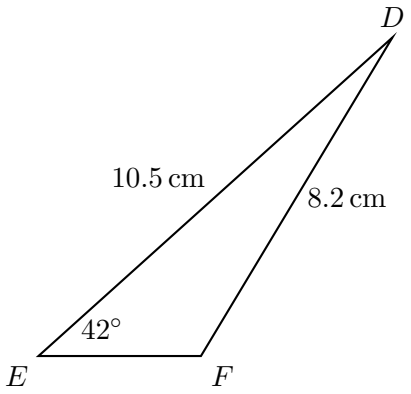
(b) Calculate the length of AC.

$AC =$ ..... cm **[3]**

(c) A point D lies outside the triangle ABC such that $CD = 6.4\,\text{cm}$ and angle ACD is acute.
The area of triangle ACD is $8.5\,\text{cm}^2$.
Calculate the size of angle ACD.

..... ° **[2]**

31 The diagram shows triangle DEF.
$DE = 10.5$ cm, $DF = 8.2$ cm and angle $DEF = 42°$.
Angle DFE is an obtuse angle.



(a) Calculate angle DFE.

Angle $DFE =$ ..... ° **[3]**

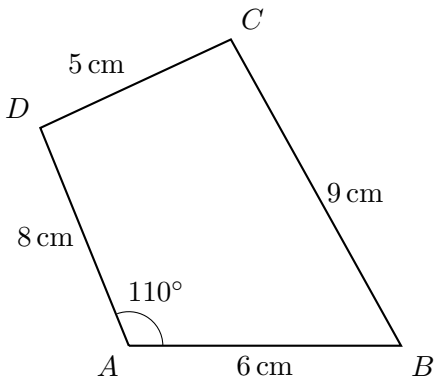
(b) Calculate the length of EF.

$EF =$ ..... cm **[3]**

(c) Calculate the shortest distance from D to the line EF.

..... cm **[2]**

32 The diagram shows a quadrilateral $ABCD$.
$AB = 6$ cm, $AD = 8$ cm, $DC = 5$ cm, and $BC = 9$ cm.
Angle $DAB = 110°$.



NOT TO SCALE

(a) Calculate the length of the diagonal BD.

..... *cm* **[3]**

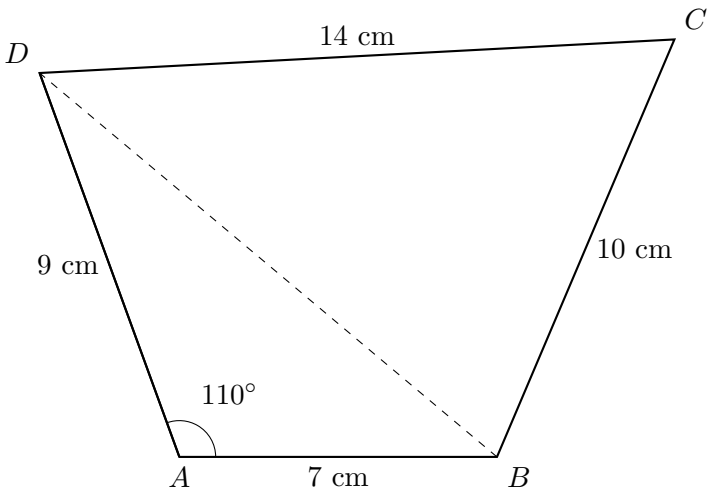
(b) Calculate the size of angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

33 The diagram shows a quadrilateral $ABCD$.
$AB = 7$ cm, $AD = 9$ cm, $BC = 10$ cm and $CD = 14$ cm.
Angle $BAD = 110°$.



(a) Calculate the length of BD.

..... cm **[3]**

(b) Calculate angle BCD.

..... ° **[3]**

(c) Calculate the total area of the quadrilateral $ABCD$.

..... cm^2 **[3]**

34 The diagram shows a quadrilateral $ABCD$.
$AB = 7.4$ cm, $BC = 9.1$ cm, and $CD = 10.5$ cm.
Angle $ABC = 112°$ and angle $CAD = 43°$.

(a) Calculate the length of AC.

..... cm **[3]**

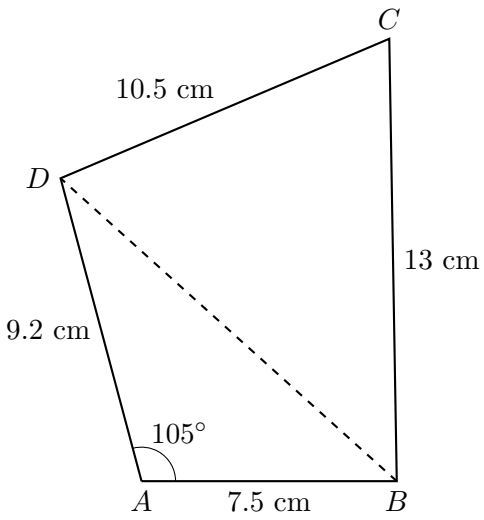
(b) Calculate the acute angle ADC.

..... ° **[3]**

(c) Calculate the shortest distance from D to the line AC.

..... cm **[3]**

35 The diagram shows a quadrilateral $ABCD$.
$AB = 7.5$ cm, $AD = 9.2$ cm and angle $BAD = 105°$.
$BC = 13$ cm and $CD = 10.5$ cm.



(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

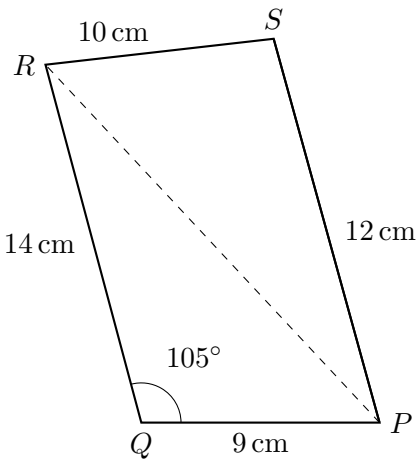
(b) Calculate angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[3]**

36 The diagram shows a quadrilateral $PQRS$.
$PQ = 9$ cm, $QR = 14$ cm, and angle $PQR = 105°$.
$PS = 12$ cm and $RS = 10$ cm.



NOT TO SCALE

(a) Calculate the length of the diagonal PR.

$PR =$ ..... cm **[3]**

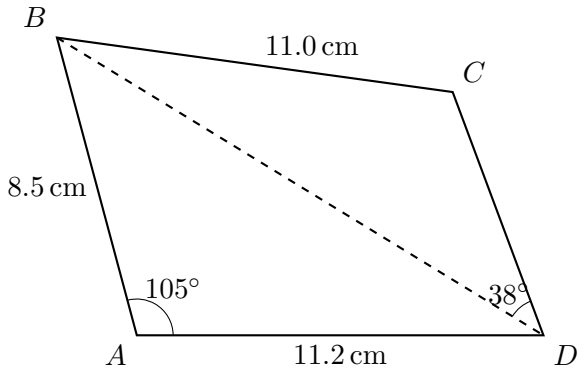
(b) Calculate angle PSR.

Angle $PSR =$ ..... ° **[4]**

(c) Calculate the total area of the quadrilateral $PQRS$.

Area = ... cm^2 **[3]**

37 The diagram shows a quadrilateral $ABCD$.
$AB = 8.5\,\text{cm}$, $AD = 11.2\,\text{cm}$ and angle $BAD = 105°$.
$BC = 11.0\,\text{cm}$ and angle $BDC = 38°$.
Angle BCD is obtuse.



NOT TO SCALE

(a) Calculate the length of BD.

$BD =$ ..... cm **[3]**

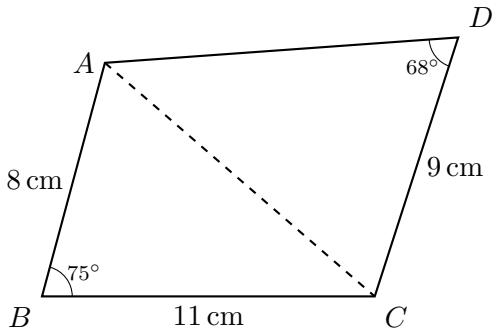
(b) Calculate the size of angle BCD.

Angle $BCD =$ ..... ° **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

38 The diagram shows a quadrilateral $ABCD$.
$AB = 8$ cm, $BC = 11$ cm and $CD = 9$ cm.
Angle $ABC = 75°$ and angle $ADC = 68°$.



(a) Calculate the length of AC.

$AC =$ ..... cm **[3]**

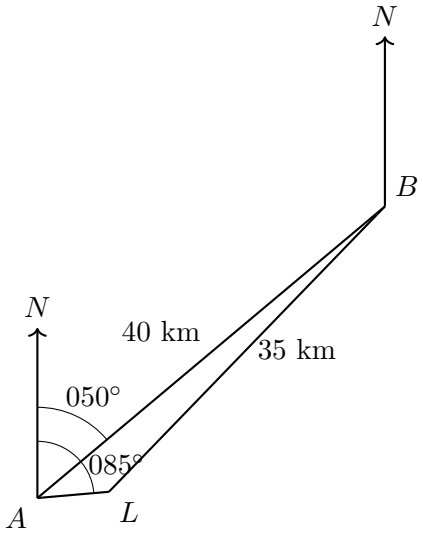
(b) Calculate the size of angle CAD.

Angle $CAD =$ ..... $°$ **[3]**

(c) Calculate the area of the quadrilateral $ABCD$.

Area = ... cm^2 **[4]**

39 Ship A and Ship B are at sea.
Ship B is 40 km from Ship A on a bearing of 050°.
A lighthouse L is 35 km from Ship B.
The bearing of L from A is 085°.
Given that the angle ALB is obtuse:



(a) Calculate the obtuse angle ALB.

..... ° **[4]**

(b) Calculate the distance AL.

..... km **[3]**

(c) Calculate the bearing of L from B.

..... ° **[3]**